

Project 2 - Enterprise Customer Data Engineering Platform

PROJECT 2

|

|— datasets

|

|— customers_csv.csv

|

|— customers_excel.xlsx

|

|— customers_json.json

|

|— api_customers.json

|

|— cleaned_data

|

|— customer_master.csv

|

|— customer_features.csv

|

|— customer_master_dataset.csv

|

|— customer_segments.csv

|

|— segment_summary.csv

|

|— analytics_dataset.csv

|

|— ml_ready_dataset.csv

|

|— reports

|

|— validation_report.csv

|

|— correlation_report.csv

|

|— business_insights.txt

|

|— data_quality_report.xlsx

|

|— logs

```
|  └─ processing.log
|
|  └─ sql
|  └─ customer_database.sql
|
|  └─ src
|  └─ load_data.py
|  └─ validate.py
|  └─ clean_data.py
|  └─ feature_engineering.py
|  └─ merge_data.py
|  └─ statistics.py
|  └─ segmentation.py
|  └─ mysql_store.py
|  └─ export.py
|  └─ pipeline.py
|
|  └─ main.py
|  └─ requirements.txt
└─ README.md
```

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college_db
- company_db
- cust_db
- custhumanlys_db
- custpurchist_db
- custsegsdb_db
- emp_db
- empdept_db
- employee_attrition
- employee_db
- empperf_db
- emprank_db
- medical_db
- placement_db
- prodsale_db
- retail_hi

Administration Schemas

Information

No object selected

Result Grid

Filter Rows: Export: Wrap Cell Content: 15

Tables_in_customerdb

customers

SQL File 17*

```

1 CREATE DATABASE CustomerDB;
2
3 USE CustomerDB;
4
5 CREATE TABLE customers (
6     CustomerID INT PRIMARY KEY,
7     Name VARCHAR(100),
8     Gender VARCHAR(20),
9     Age INT,
10    City VARCHAR(100),
11    State VARCHAR(100),
12    Country VARCHAR(100),
13    Phone VARCHAR(20)
14 );

```

Object Info Session Result 1 x Read Only

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college_db
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- prodsale_db
- retail_hi

Administration Schemas

Information

No object selected

Result Grid

Filter Rows: Export: Wrap Cell Content: 15

Tables_in_customerdb

customers

SQL File 17*

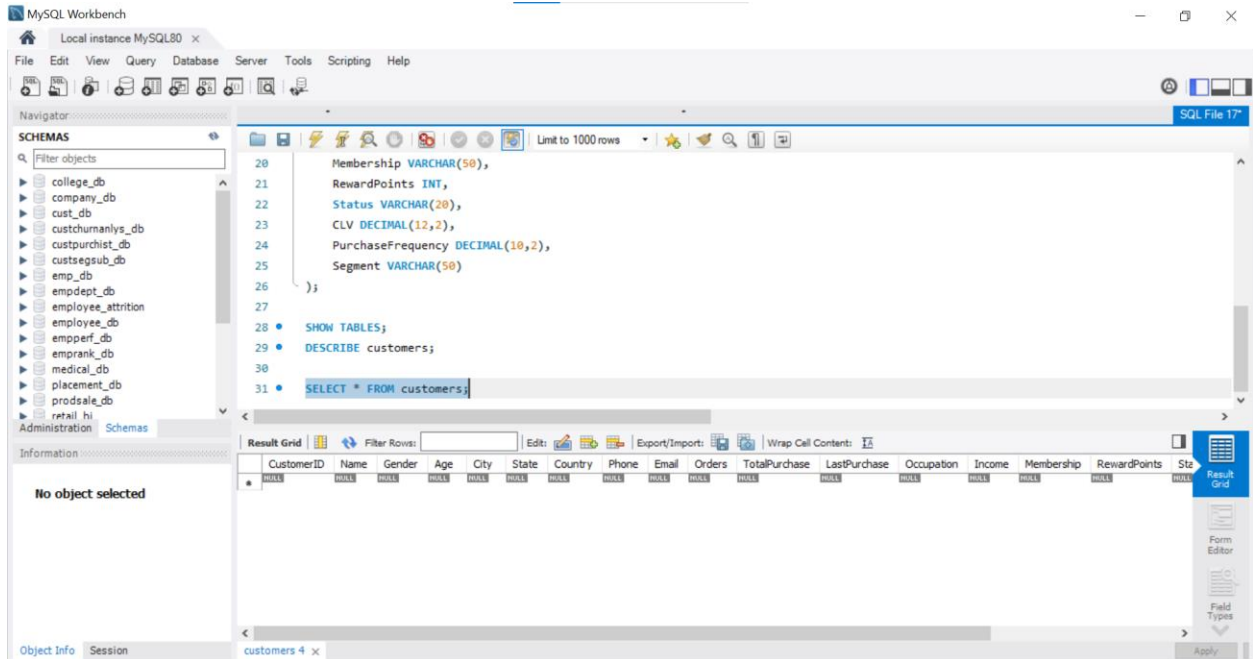
```

18 Occupation VARCHAR(100),
19 Income DECIMAL(12,2),
20 Membership VARCHAR(50),
21 RewardPoints INT,
22 Status VARCHAR(20),
23 CLV DECIMAL(12,2),
24 PurchaseFrequency DECIMAL(10,2),
25 Segment VARCHAR(50)
26 );
27
28 SHOW TABLES;
29 DESCRIBE customers;

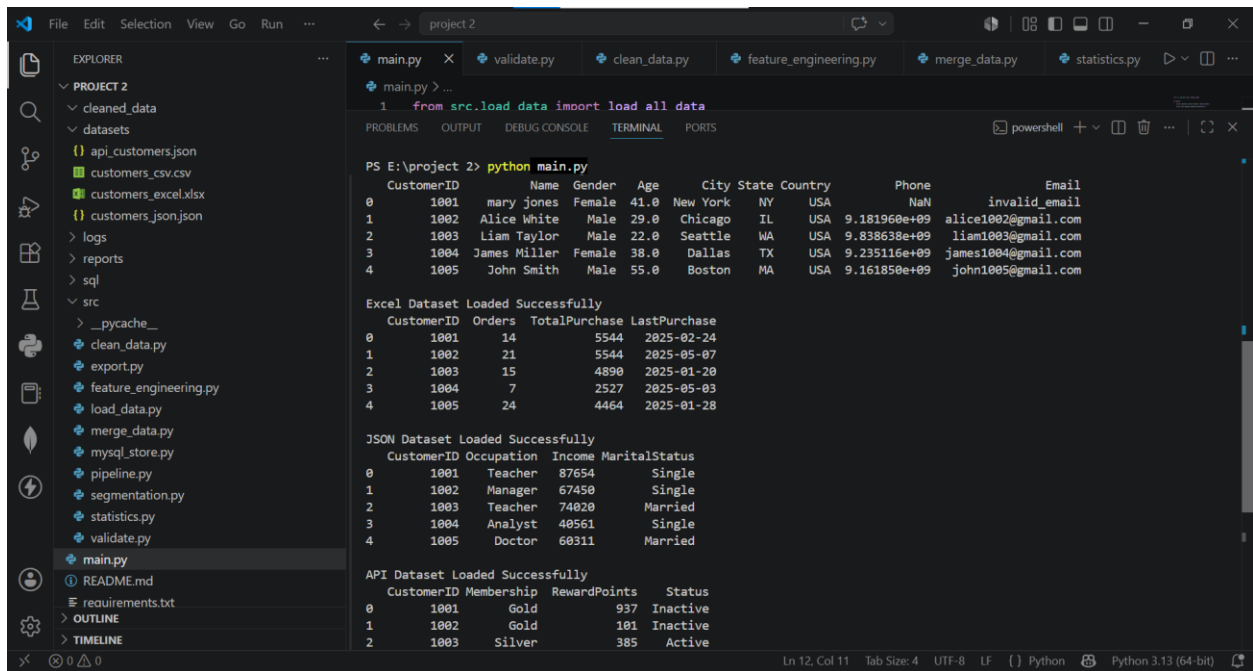
```

Field	Type	Null	Key	Default	Extra
CustomerID	int	NO	PRI		
Name	varchar(100)	YES			
Gender	varchar(20)	YES			
Age	int	YES			
City	varchar(100)	YES			
State	varchar(100)	YES			
Country	varchar(100)	YES			
Phone	varchar(20)	YES			
Email	varchar(100)	YES			
Orders	int	YES			

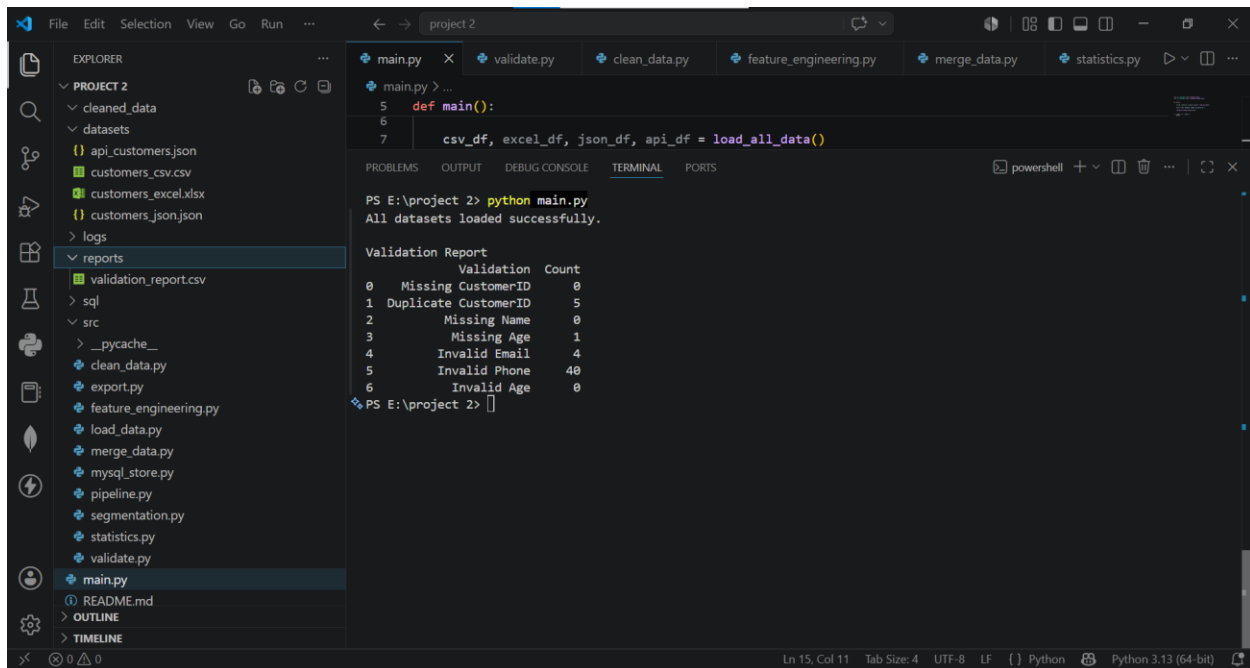
Object Info Session Result 1 x Read Only



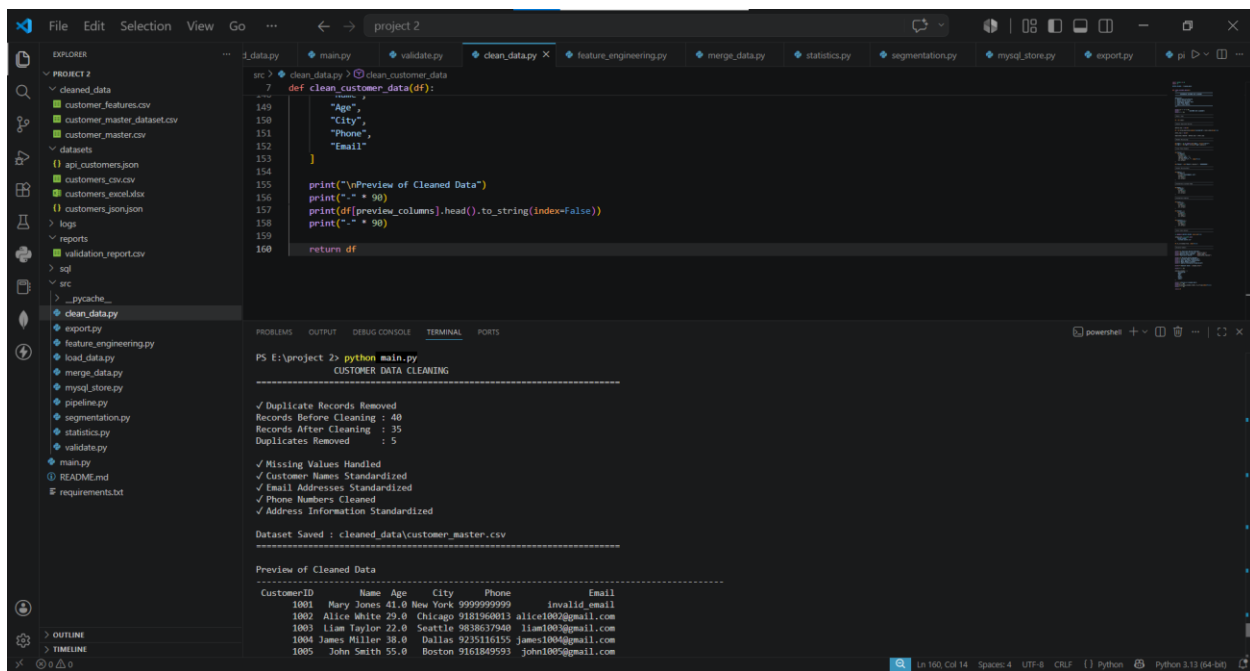
Requirement 2:



Requirement 3



Requirement 4,5,6



Requirement 7

```

File Edit Selection View Go ... project 2
EXPLORE PROJECT 2
  cleaned_data
  customer_features.csv
  customer_master_dataset.csv
  customer_master.csv
  datasets
  api_customers.json
  customers.csv
  customers.csv.csv
  customers.json
  customers.json.json
  logs
  reports
  validation_report.csv
  __pycache__
  clean_data.py
  export.py
  feature_engineering.py
  load_data.py
  merge_data.py
  mysql_store.py
  pipeline.py
  requirements.txt
  README.md
  requirements.txt

src > merge_data.py > merge_customer_data
7 def merge_customer_data(customer_df, purchase_df, json_df, api_df):
8     .head()
9     .to_string(index=False)
10
11     print("-- * * 120)
12
13     return merged_df
14
15 PS E:\project 2> python main.py
16
17 =====
18 FEATURE ENGINEERING STARTED
19
20 ✓ Customer Dataset Merged Successfully
21 ✓ Feature Engineering Completed
22
23 Feature Name      Status
24 -----
25 Address           ✓
26 PurchasedFrequency ✓
27 Customer Lifetime Value ✓
28 LoyaltyCustomer   ✓
29 PurchaseCategory  ✓
30
31 Total Customers Processed : 35
32 Total Columns Available : 17
33 Dataset Saved : cleaned_data\customer_features.csv
34
35 Preview (First 5 Records)
36
37 CustomerID  Name Gender Age City State Country Phone Email Orders TotalPurchase LastPurchase AgeGroup PurchasedFrequency CLV LoyaltyCustomer PurchaseCategory
38 1001 Mary Jones Female 41.0 New York NY USA 9999999999 invalid_email 14 5544.0 2025-02-24 Senior 1.17 77616.0 Yes High Value
39 1002 Alice White Male 29.0 Chicago IL USA 9181908011.0 alice1002@gmail.com 21 5544.0 2025-05-07 Adult 1.75 116424.0 Yes High Value
40 1003 Liam Taylor Male 22.0 Seattle WA USA 9836379948.0 liam1003@gmail.com 15 4098.0 2025-01-20 Young 1.25 73358.0 Yes Medium Value
41 1004 James Miller Female 38.0 Dallas TX USA 9235116155.0 james1004@gmail.com 7 2527.0 2025-05-03 Adult 0.50 17689.0 No Medium Value
42 1005 John Smith Male 55.0 Boston MA USA 9181949993.0 john1005@gmail.com 24 4464.0 2025-01-28 Senior 2.00 187136.0 Yes Medium Value
43
44 =====
45 CUSTOMER DATA MERGING STARTED
46
47 ✓ Customer Dataset + Purchase Dataset Merged
48 ✓ Demographic Dataset Merged
49 ✓ API Dataset Merged
50
51 =====
52 DATA MERGING COMPLETED
53
54 Total Records : 35
55 Total Columns : 26
56 Dataset Saved : cleaned_data\customer_master_dataset.csv
57
58 Merged Datasets
59
60 ✓ Customer CSV
61 ✓ Purchase Batch
62 ✓ API Customer 350h
63
64 Preview of Master Dataset
65
66 CustomerID  Name Age City Orders TotalPurchase Income Membership RewardPoints CLV PurchaseCategory
67 1001 Mary Jones 41.0 New York 14 5544.0 67005.0 Gold 927.0 77616.0 High Value
68 1002 Alice White 29.0 Chicago 21 5544.0 67450.0 Gold 181.0 116424.0 High Value
69 1003 Liam Taylor 22.0 Seattle 15 4098.0 38208.0 Silver 35.0 73358.0 Medium Value
70 1004 James Miller 38.0 Dallas 7 2527.0 48951.0 Silver 38.0 17689.0 Medium Value
71 1005 John Smith 55.0 Boston 24 4464.0 68111.0 Silver 334.0 187136.0 Medium Value
72
73 PS E:\project 2>

```

Requirement 8

```

File Edit Selection View Go ... project 2
EXPLORE PROJECT 2
  cleaned_data
  customer_features.csv
  customer_master_dataset.csv
  customer_master.csv
  datasets
  api_customers.json
  customers.csv
  customers.csv.csv
  customers.json
  customers.json.json
  logs
  reports
  validation_report.csv
  __pycache__
  clean_data.py
  export.py
  feature_engineering.py
  load_data.py
  merge_data.py
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  requirements.txt

src > merge_data.py > merge_customer_data
7 def merge_customer_data(customer_df, purchase_df, json_df, api_df):
8     .head()
9     .to_string(index=False)
10
11     print("-- * * 120)
12
13     return merged_df
14
15 PS E:\project 2> python main.py
16
17 =====
18 CUSTOMER DATA MERGING STARTED
19
20 ✓ Customer Dataset + Purchase Dataset Merged
21 ✓ Demographic Dataset Merged
22 ✓ API Dataset Merged
23
24 =====
25 DATA MERGING COMPLETED
26
27 Total Records : 35
28 Total Columns : 26
29 Dataset Saved : cleaned_data\customer_master_dataset.csv
30
31 Merged Datasets
32
33 ✓ Customer CSV
34 ✓ Purchase Batch
35 ✓ API Customer 350h
36
37 Preview of Master Dataset
38
39 CustomerID  Name Age City Orders TotalPurchase Income Membership RewardPoints CLV PurchaseCategory
40 1001 Mary Jones 41.0 New York 14 5544.0 67005.0 Gold 927.0 77616.0 High Value
41 1002 Alice White 29.0 Chicago 21 5544.0 67450.0 Gold 181.0 116424.0 High Value
42 1003 Liam Taylor 22.0 Seattle 15 4098.0 38208.0 Silver 35.0 73358.0 Medium Value
43 1004 James Miller 38.0 Dallas 7 2527.0 48951.0 Silver 38.0 17689.0 Medium Value
44 1005 John Smith 55.0 Boston 24 4464.0 68111.0 Silver 334.0 187136.0 Medium Value
45
46 PS E:\project 2>

```

Requirements 11, 17

The screenshot shows a Visual Studio Code editor with a project named 'project 2'. The Explorer pane on the left shows a file structure with folders for 'cleaned_data', 'datasets', 'logs', 'reports', 'src', and 'sql'. The 'main.py' file is selected and open in the editor. The code in 'main.py' includes a 'def main()' function that calls 'generate_statistics(master_df)'. The output window on the right shows the results of the script execution, including a 'CUSTOMER STATISTICAL ANALYSIS' section with various metrics and a 'Membership Distribution' table.

```
def main():  
    # Step 6 : Statistical Analysis  
    # =====  
    generate_statistics(master_df)
```

PS E:\project 2> python main.py

CUSTOMER STATISTICAL ANALYSIS

Total Customers : 35
Average Age : 37.64
Average Income : \$43917.26
Average Purchase : \$3682.91
Average CLV : \$62645.83
Total Sales : \$128902.0

Membership Distribution

Membership

None 25
Silver 6
Gold 3
Bronze 1
Name: count, dtype: int64

Top Customer Cities

City

New York 9
Dallas 9
Chicago 6
Boston 6
Seattle 5
Name: count, dtype: int64

✓ Correlation Report Saved
✓ Business Insights Generated

ETL PIPELINE EXECUTED SUCCESSFULLY

PS E:\project 2>

Requirement 10, 17

The screenshot shows the same Visual Studio Code editor with 'project 2'. The 'main.py' file is open, and the code is being executed. The output window shows the results of the script, including a 'CUSTOMER SEGMENTATION' section with a table of customer counts by segment and a list of files created.

```
def main():  
    # Step 6 : Statistical Analysis  
    # =====  
    generate_statistics(master_df)
```

PS E:\project 2> python main.py

Name: count, dtype: int64

✓ Correlation Report Saved
✓ Business Insights Generated

CUSTOMER SEGMENTATION

Customer Segmentation Completed

Segment CustomerCount
Bronze 16
Platinum 10
Gold 5
Silver 4

Files Created

✓ customer_segments.csv
✓ segment_summary.csv

ETL PIPELINE COMPLETED SUCCESSFULLY

PS E:\project 2>

Requirements 13, 19, and 20

The screenshot shows a Visual Studio Code editor with a project named 'project 2'. The Explorer sidebar on the left displays a file structure with folders like 'cleaned_data', 'datasets', 'logs', 'reports', 'sql', and 'src'. The 'src' folder contains several Python files, including 'mysql_store.py' which is currently selected. The main editor window shows the code for 'mysql_store.py', specifically the 'store_to_mysql' function. The output pane at the bottom displays the results of running 'python main.py', showing successful data storage in MySQL.

```
src > mysql_store.py > store_to_mysql
118 | print("a" * 70)

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\project 2> python main.py
✓ customer_segments.csv
✓ segment_summary.csv

=====
STORING DATA INTO MYSQL
=====
✓ Records Inserted : 35
✓ Duplicate Records Skipped : 0
Data successfully stored in MySQL.

=====
ETL PIPELINE COMPLETED SUCCESSFULLY
=====

PS E:\project 2>
```

Requirements 14 and 18

The screenshot shows the same Visual Studio Code editor with 'project 2'. The Explorer sidebar is visible, and the 'main.py' file is now selected in the 'src' folder. The main editor window shows the code for 'main.py', including the 'main()' function. The output pane at the bottom displays the results of running 'python main.py', showing successful export of datasets and creation of files.

```
main.py > ...
32 | def main():
103 |
104 |     print("\n" + "=" * 80)
105 |     print("          ETL PIPELINE COMPLETED SUCCESSFULLY")
106 |     print("=" * 80)
107 |
108 |     return segmented_df
109 |
110 |
111 | if __name__ == "__main__":
112 |     main()

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\project 2> python main.py

=====
EXPORTING DATASETS
=====
✓ Analytics Dataset Exported
✓ ML Ready Dataset Exported
✓ Data Quality Report Generated

Files Created
-----
analytics_dataset.csv
ml_ready_dataset.csv
data_quality_report.xlsx

=====
ETL PIPELINE COMPLETED SUCCESSFULLY
=====

PS E:\project 2>
```


VS Code interface showing a Python project named "project 2". The Explorer sidebar on the left displays the project structure, including files like `cleaned_data`, `datasets`, `api_customers.json`, `customers_csv.csv`, `customers_excel.xlsx`, `customers_json.json`, `logs`, `processing.log`, `reports`, `business_insights.txt`, `correlation_report.csv`, `data_quality_report.xlsx`, `validation_report.csv`, `sql`, `src`, `_pycache_`, `clean_data.py`, `export.py`, `feature_engineering.py`, `load_data.py`, `merge_data.py`, `mysql_store.py`, `pipeline.py`, `segmentation.py`, `statistics.py`, `validate.py`, `main.py`, `README.md`, and `requirements.txt`.

The main editor displays the `main.py` file with the following code:

```
1 """
2 =====
3 Enterprise Customer Data Engineering Platform
4 =====
5 """
6
7 from src.pipeline import run_pipeline
8
9
10 if __name__ == "__main__":
11     run_pipeline()
```

The TERMINAL panel at the bottom shows the output of running `python main.py` in a PowerShell session:

```
PS E:\project 2> python main.py
=====
EXPORTING DATASETS
=====
✓ Analytics Dataset Exported
✓ ML Ready Dataset Exported
✓ Data Quality Report Generated

Files Created
-----
analytics_dataset.csv
ml_ready_dataset.csv
data_quality_report.xlsx

=====
ETL PIPELINE EXECUTED SUCCESSFULLY
=====
PS E:\project 2>
```

The status bar at the bottom indicates the current position is Line 11, Column 19, with 4 tabs open, using UTF-8 encoding, LF line endings, Python 3.13 (64-bit).